

# CELL BIOLOGY – ANIMAL CELL STRUCTURE

## KEY VOCABULARY

**Nucleus:** controls the cell and contains the genetic material (DNA).

**Cytoplasm:** jelly-like fluid where most reactions happen.

**Mitochondria:** site of aerobic respiration; releases energy.

**Ribosomes:** where protein synthesis takes place.

## COMMON MISCONCEPTIONS

~~Animal cells have a cell wall.~~

✓ Only plant, fungal and bacterial cells have walls.

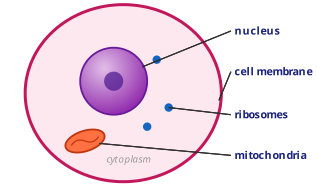
~~The nucleus makes energy.~~

✓ Mitochondria release energy in respiration.

~~All cells are the same size and shape.~~

✓ Cells are specialised for their function.

## A TYPICAL ANIMAL CELL



The **cell membrane** controls what enters and leaves the cell.

## 1 LABEL THE CELL

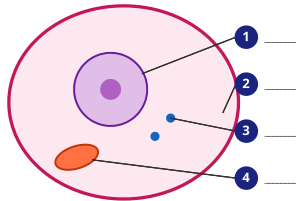
Name parts **1-4** on the animal cell using the word bank. **[4 marks]**

nucleus

cell membrane

mitochondria

ribosomes



## 2 STATE THE FUNCTION

State the **function** of the nucleus and of the mitochondria. **[2 marks]**

## 3 ANIMAL VS PLANT

Name **two** structures found in a plant cell but not in an animal cell. **[2 marks]**

## 4 EUKARYOTIC CELLS

**Explain** why an animal cell is described as eukaryotic. **[2 marks]**

## 5 MAGNIFICATION

An image is 30 mm wide and the real cell is 0.06 mm wide. **Calculate** the magnification. **[2 marks]**

Answer: \_\_\_\_\_ ×

## 6 SPECIALISED CELLS

A sperm cell has many mitochondria. **Suggest** why. **[2 marks]**